CariSurg MedTech Pathways - Week 0

Day 5: Unconsidered Metrics

Vital Sign: Oxygen Saturation (SpO_2)

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Peripheral oxygen saturation (SpO_2) is a non-invasive measure of how much haemoglobin in the blood is carrying oxygen, expressed as a percentage and recorded using a pulse oximeter placed on the fingertip or earlobe (*Blood Oxygen Level: What It Is & How To Increase It*, 2022). In a healthy adult, a normal SpO_2 reading falls between 95% and 98%, while readings below 94% are considered clinically significant and anything below 90% constitutes a medical emergency requiring urgent intervention (Piraino et al., 2022). The Mercer General triage dataset included FiO_2, which records how much supplemental oxygen a patient is being given, but does not include SpO_2, which records whether that oxygen is actually reaching the blood effectively. This is a meaningful gap because FiO_2 alone cannot confirm adequate oxygenation; a patient on 100% oxygen may still be critically hypoxic if their lungs are too damaged to transfer it. Studies prove that the lower a hospital patient's blood oxygen drops, the higher their risk of dying, with even mild drops (90% to 93%) showing a much higher death rate than healthy levels (above 95%) (Graham et al., 2024). In triage, low SpO_2 has been identified as one of the vital signs most strongly associated with mortality, alongside abnormal blood pressure and respiratory rate (Simbawa et al., 2021). Because the dataset lacks blood oxygen levels (SpO_2), any model built from it can only track the oxygen treatment delivered, not whether the patient's body is actually absorbing it.

References

Blood Oxygen Level: What It Is & How To Increase It. (2022, February 18). Cleveland Clinic. https://my.clevelandclinic.org/health/diagnostics/22447-blood-oxygen-level

Graham, H. R., King, C., Duke, T., Ahmed, S., Baqui, A. H., Colbourn, T., Falade, A. G., Hildenwall, H., Hooli, S., Kamuntu, Y., Subhi, R., & McCollum, E. D. (2024). Hypoxaemia and risk of death among children: rethinking oxygen saturation, risk-stratification, and the role of pulse oximetry in primary care. The Lancet. Global health, 12(8), e1359–e1364. https://doi.org/10.1016/S2214-109X(24)00209-2

Piraino, T., Madden, M., Roberts, K. J., Lamberti, J., Ginier, E., & Strickland, S. L. (2022). AARC Clinical Practice Guideline: Management of Adult Patients With Oxygen in the Acute Care Setting. Respiratory care, 67(1), 115–128. https://doi.org/10.4187/respcare.09294

Simbawa, J. H., Jawhari, A. A., Almutairi, F., Almahmoudi, A., Alshammrani, B., Qashqari, R., & Alattas, I. (2021). The Association Between Abnormal Vital Signs and Mortality in the Emergency Department. Cureus, 13(12), e20454. https://doi.org/10.7759/cureus.20454